

NATALIE CHMURA

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EDUCATION

The Ohio State University

Honors BS in Computer and Information Science, Minor: Japanese

Scholarship: Choose Ohio First; Excellence Fellowship in Computing

May 2025

Columbus, Ohio

SKILLS

Programming Languages: C, C#, C++, Rust, Java, Python, HTML, CSS, Asm (x86), $\text{\LaTeX}$, JS, TS, Scheme, SQL

Software and Frameworks: ROS 2 (Foxy and Humble), Git, SVN, QuartusPrime, Angular, React, Maven, SpringBoot

Interpersonal: Group Leadership, Active Listening, Program Development, Resource Management

PROJECTS

Polyomino Generator: | Python, Tkinter

- Visualization tool to help identify and generate polyominoes to aid conceptualizing a currently unsolved problem in mathematics.

Interpreter: | Java

- Interpreter for the "Core" language, with a helper Scanner and Parse Tree.

Compiler: | Java

- Built a compiler for a subset of the C language with added control flow analysis.

2D Video Game: | C#, .Net, VisualStudio, Monogame

- Worked in a small team using agile methodology to develop a Super Mario Bros clone with added features and game modes over the course of a semester.

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Network Based Computing Laboratory

August 2024 - May 2025

Columbus, Ohio

- Wrote benchmark scripts to test message-passing library optimizations, and optimal configuration parameters.
- Presented poster about my work at the 2025 Ohio Conference for Women in Computing (OCWIC)
- Planning and testing generalized auto tuning functionality
- Software and Languages utilized: C, Bash, YAML, MPI, MVAPICH, Git

Undergraduate Research Assistant

Artificially Intelligent Manufacturing Systems Lab

May 2023 - February 2024

Columbus, Ohio

- Implemented path planning algorithm to help the robotic end effector navigate convex path areas.
- Wrote wrappers to provide reach analysis data points.
- Designed a program that generated and converted Point Cloud Data into different formats
- Ported existing work-cells and packages from ROS 1 into ROS 2
- Software and Languages utilized: ROS 2 (Foxy, Humble), C++, Python, CMake, Bash, XML, URDF, Git

PRESENTATIONS

MPI Benchmark Automation in High Performance Computing

Ohio Conference for Women in Computing

February 21, 2025

Huron, Ohio

- Discussed design philosophy and impact of automating the process used to run and collect benchmarking data.

WORK EXPERIENCE

Technology Intern

Chemical Abstracts Services

May 2024 - August 2024

Columbus, Ohio

- Redesigned internal web-app based tool from using AngularJS 1.7 to Angular 18, transferred architecture to a more modular, component-based design, QA Tested newer and legacy systems
- QA Testing, validation of both newer and legacy systems
- Software/ Languages Used: JS, TS, Java, Perl, Bash, AngularJS, Angular Cli, SpringBoot, Maven, Git

Undergraduate Teaching Assistant, Project: Interactive Game Systems

Ohio State University, College of Engineering

January 2024 - May, 2025

Columbus, Ohio

- Intensive group project involving design, development, and documentation of an interactive software system, a 2D interactive game; communication skills emphasized; builds programming maturity.

- Software/ Languages Taught: C#, .Net, Visual Studio, Monogame, Git

Undergraduate Teaching Assistant, Data Structures and Algorithms

January, 2023 – May, 2024

Ohio State University, College of Engineering

Columbus, Ohio

- Design/analysis of algorithms and data structures
- Assisted professor in lecture, proctoring exams, and grading homework

Undergraduate Teaching Assistant, Software Development and Design

August, 2022 – May 2023

Ohio State University, College of Engineering

Columbus, Ohio

- Data representation using hashing, search trees, and linked data structures; algorithms for sorting; using trees for language processing; component interface design; best practices in Java
- Software/ Languages Taught: HTML, CSS, Java, Eclipse IDE, SVN
- Assisted professor and fellow TA with twice weekly lab sessions and held twice weekly office hours

Student Manager and Student Community Lead

September, 2020 – February 2023

Ohio State University, Dining Services

Columbus, Ohio

- Met for weekly leadership and development sessions, assisted organizing and running monthly community events
- Interviewed, ran orientation shifts for and trained new employees
- Handled routine complications and maintained facility cleanliness and customer satisfaction through stocking, serving, and assisting fellow employees

Line Server

May 2022 - August 2022

Chipotle Mexican Grill

Parma, Ohio

- Worked in front and back of house, prepping food and serving in person and mobile customers.

Line Server

May 2021 - July 2021

Panera Bread

Broadview Heights, Ohio

- Worked in front and back of house, prepping food and serving customers, both in bakery and line

Key Holder

May 2019 - August 2021

Pulp Juice and Smoothie

Independence, Ohio

- Ensured customer satisfaction and facility cleanliness!
- Held locking/unlocking duties, including shift end drawer counting, and new employee training.

EXTRACURRICULARS

Open Source Club

2025

Member

Ohio State University

- Group discussions and presentations on topics in the open source community.
- Winner of OSC's 2025 template-a-thon (all submissions available on LibreOffice impress templates page!)

Association of Computing Machinery Women's Chapter

Autumn 2021 – May 2025

Member

Ohio State University

- Engaged with fellow students and the community at large in order to improve the working and learning environment of technology for its women and nonbinary members.
- Engaged in outreach events fostering interest and accessibility in computing education for grades K-12
- Mentored underclassmen for 2+ years
- Attended 2023 OCWIC conference to connect with other Ohio School ACMW chapter members

Bread Club

Autumn 2020 - Spring 2023

Member and Treasurer

Ohio State University

- Resumed meetings after online only participation, managing membership fees, kitchen reservation fees and weekly ingredient costs for 30+ members.

Yugioh Club

Autumn 20201 - Spring 2023

Member

Ohio State University

- Met weekly to participate in the noncompetitive play and inner-club tournaments.

VOLUNTEERING

ACM-W: Assisted in outreach programs including hosting coding camps and local events to foster an interest in computing in grades K-12!

Tutoring: Intro to Java, Intro to C++, Data Structures and Algorithms